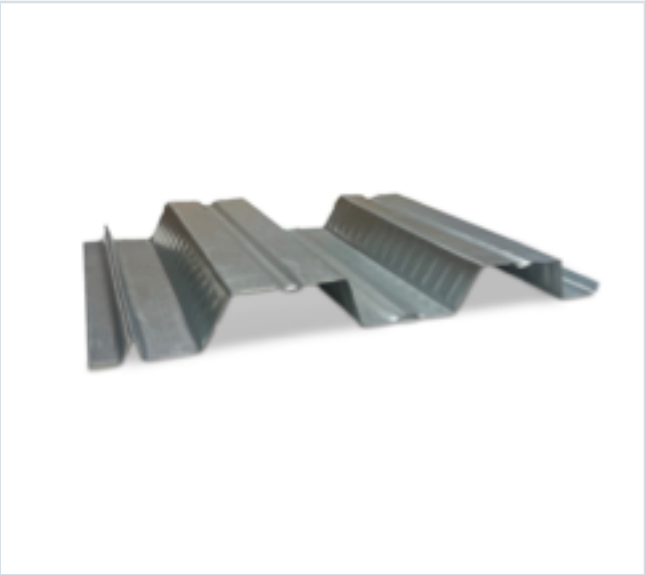


DECKING PRODUCT SUBMITTAL

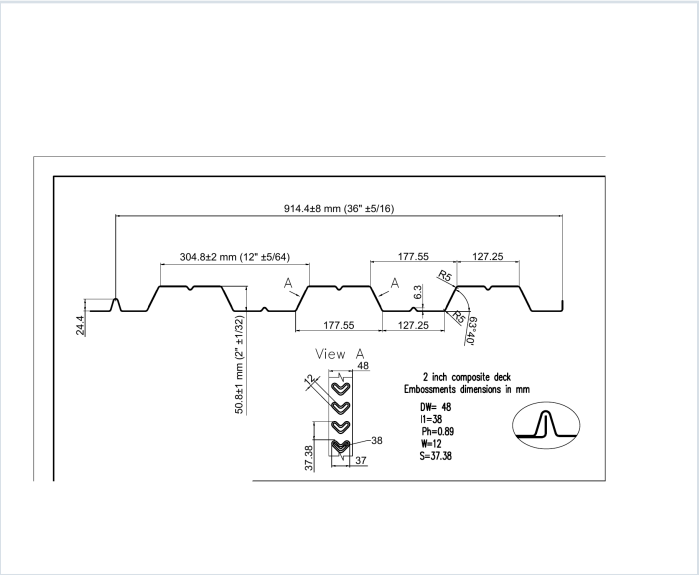
2CD-FY50-20-G60

PRODUCT	GRADE	GAUGE	COATING
2" Composite Deck	Grade 50	20 ga	G60

PRODUCT VIEW



PROFILE DRAWING



SECTION PROPERTIES · 20 GAUGE

Thickness (in.)	Weight (psf)	Fy (ksi)	S+ (in ³ /ft)	S- (in ³ /ft)	M+ (in-lb/ft)	M- (in-lb/ft)	I+ (in ⁴ /ft)	I- (in ⁴ /ft)
0.0358	1.9	50	0.329	0.336	9860	10060	0.393	0.373

SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 54, 55, 56, 57, 58, 59, 60, 61
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

PUBLISHED PERFORMANCE DATA

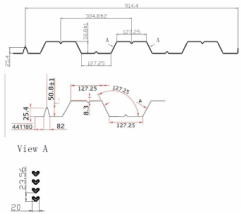
2CD-FY50-20-G60 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 54

GRADE 50 STEEL



Section Properties

Gage	Design Thickness (inches)	Weight (pcf)	E _x (ksi)	S _x (in ³)	S _y (in ³)	ASD (ft + 1/8")	M _y (ft + 1/8")	I _y (in ⁴)	I _x (in ⁴)
22	0.0205	16	50	0.247	0.254	2427	2427	0.31	0.290
20	0.0358	19	50	0.328	0.336	3860	3860	0.393	0.373
18	0.0474	21	50	0.408	0.500	14700	14700	0.548	0.530
16	0.0598	23	50	0.465	0.664	19101	19101	0.703	0.683

Note: All section properties and ASD based strengths are calculated in accordance with ANSI/AISC 360-10, AISI S100-2017, and AISI S100-2016.

Shear and Web Crippling

Gage	V _y /D (lbs/ft)	Web Crippling (R _y /D), lbs/ft One Flange Loading End Bearing			Web Crippling (R _y /D), lbs/ft One Flange Loading Interior Bearing		
		2"	3"	4"	2"	3"	4"
22	388	434	476	529	627	709	779
20	278	590	779	749	904	1018	1113
18	365	986	1123	1239	1532	1752	1865
16	460	1507	1707	1875	2367	2631	2854

Note: All section properties and ASD based strengths are calculated in accordance with ANSI/AISC 360-10, AISI S100-2017, and AISI S100-2016.

STEEL CATALOG PAGE 55

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	137	133	77	61	49	41	34	29	25	22	19
	20	181	134	103	81	66	54	44	36	30	26	23
	18	273	203	154	121	95	81	68	58	50	44	38
	16	358	263	203	159	129	106	89	76	66	57	50
Double	22	131	113	79	63	51	42	35	30	26	23	20
	20	196	137	105	83	67	55	47	40	34	30	26
	18	277	204	156	123	100	82	69	59	51	44	39
	16	357	262	203	159	129	106	89	76	66	57	50
Triple	22	176	129	99	78	63	52	44	37	32	28	25
	20	223	171	131	105	84	69	58	50	43	37	33
	18	346	254	195	154	125	103	87	74	64	55	49
	16	446	338	251	198	161	133	112	95	82	71	63

Notes:
1. All section properties and ASD (2 + 1/8") uniform loads are calculated in accordance with ANSI/AISC 360-10, AISI S100-2017, and AISI S100-2016.
2. Loads shown in tables are uniformly distributed superimposed loads in psi. Span lengths include end-to-end spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formula used for bearing stress limitations are: Single and Two Span: $M \leq \frac{F_y I}{8}$; Three Span: $M \leq \frac{F_y I}{10}$

Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

Uniform Superimposed Service Load that Causes L/240 Deflection (PSF)

Span	Gage	6'	7'	8'	9'	10'	11'	12'	13'	14'	15'	16'
Single	22	13	11	8	6	5	4	3	2	1	1	1
	20	13	11	8	6	5	4	3	2	1	1	1
	18	13	11	8	6	5	4	3	2	1	1	1
	16	13	11	8	6	5	4	3	2	1	1	1
Double	22	21	17	13	10	8	6	5	4	3	2	1
	20	21	17	13	10	8	6	5	4	3	2	1
	18	21	17	13	10	8	6	5	4	3	2	1
	16	21	17	13	10	8	6	5	4	3	2	1
Triple	22	24	19	14	11	9	7	6	5	4	3	2
	20	24	19	14	11	9	7	6	5	4	3	2
	18	24	19	14	11	9	7	6	5	4	3	2
	16	24	19	14	11	9	7	6	5	4	3	2

Note:
For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

PUBLISHED PERFORMANCE DATA

2CD-FY50-20-G60 · 20 GAUGE

OFFICIAL STEEL CATALOG EXCERPTS

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STEEL CATALOG PAGE 56

Normal Weight Concrete (145 pcf)				Lightweight Concrete (115 pcf)			
Maximum Unshored Clear Span				Maximum Unshored Clear Span			
400 (#200) 39 PSF	2x12x22 ga	7'-11"	9'-1"	9'-3"	2x12x22 ga	8'-6"	9'-9"
	2x12x20 ga	9'-3"	10'-6"	10'-10"	2x12x20 ga	10'-0"	11'-3"
	2x12x18 ga	10'-5"	12'-9"	12'-5"	2x12x18 ga	11'-2"	13'-9"
	2x12x16 ga	11'-4"	14'-2"	13'-4"	2x12x16 ga	12'-0"	14'-1"
450 (#250) 45 PSF	2x12x22 ga	7'-7"	8'-8"	8'-10"	2x12x22 ga	8'-2"	9'-5"
	2x12x20 ga	8'-10"	10'-0"	10'-4"	2x12x20 ga	9'-7"	10'-10"
	2x12x18 ga	9'-11"	12'-2"	12'-0"	2x12x18 ga	10'-9"	13'-3"
	2x12x16 ga	10'-11"	13'-8"	12'-10"	2x12x16 ga	11'-8"	13'-8"
500 (#300) 51 PSF	2x12x22 ga	7'-3"	8'-4"	8'-6"	2x12x22 ga	7'-11"	9'-1"
	2x12x20 ga	8'-6"	9'-7"	9'-11"	2x12x20 ga	9'-3"	10'-6"
	2x12x18 ga	9'-7"	11'-8"	11'-8"	2x12x18 ga	10'-5"	12'-5"
	2x12x16 ga	10'-6"	13'-3"	12'-6"	2x12x16 ga	11'-4"	14'-2"
550 (#350) 57 PSF	2x12x22 ga	6'-11"	8'-0"	8'-2"	2x12x22 ga	7'-7"	8'-9"
	2x12x20 ga	8'-3"	9'-3"	9'-6"	2x12x20 ga	8'-11"	10'-1"
	2x12x18 ga	9'-3"	11'-3"	11'-4"	2x12x18 ga	10'-0"	12'-4"
	2x12x16 ga	10'-1"	12'-9"	12'-2"	2x12x16 ga	11'-0"	13'-11"
600 (#400) 63 PSF	2x12x22 ga	6'-9"	7'-9"	7'-10"	2x12x22 ga	7'-5"	8'-6"
	2x12x20 ga	7'-11"	8'-11"	9'-2"	2x12x20 ga	8'-8"	9'-9"
	2x12x18 ga	8'-11"	10'-10"	11'-0"	2x12x18 ga	9'-9"	11'-10"
	2x12x16 ga	9'-9"	12'-4"	11'-10"	2x12x16 ga	10'-8"	13'-6"
650 (#450) 69 PSF	2x12x22 ga	6'-6"	7'-6"	7'-7"	2x12x22 ga	7'-2"	8'-3"
	2x12x20 ga	7'-9"	8'-7"	8'-10"	2x12x20 ga	8'-5"	9'-5"
	2x12x18 ga	8'-8"	10'-6"	10'-9"	2x12x18 ga	9'-5"	11'-6"
	2x12x16 ga	9'-6"	11'-11"	11'-7"	2x12x16 ga	10'-4"	12'-4"

Note
With crippling and clear span not shown accounted for in these tables. Required bearing should be determined based on specific span conditions.

STEEL CATALOG PAGE 61

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	375	330	290	260	230
4.5	35	400	400	400	400	395	397	384
5	39	400	400	400	400	400	370	337
5.5	44	400	400	400	400	400	400	393
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	210	180	172	157	143	131	120	110
4.5	246	239	230	191	176	160	147	138
5	304	275	250	227	208	191	175	161
5.5	354	320	291	265	242	222	204	188
6	400	367	334	304	278	255	234	216
6.5	400	400	377	344	314	288	265	244

16 ga Lightweight Concrete (115 pcf, F_c = 3,000 psi)

Slab Thickness (Inches)	Weight (pcf)	7'-0"	7'-6"	8'-0"	8'-6"	9'-0"	9'-6"	10'-0"
4	31	400	400	400	400	392	323	290
4.5	35	400	400	400	400	400	390	360
5	39	400	400	400	400	400	400	400
5.5	44	400	400	400	400	400	400	400
6	48	400	400	400	400	400	400	400
6.5	53	400	400	400	400	400	400	400

Slab Thickness (Inches)	10'-6"	11'-0"	11'-6"	12'-0"	12'-6"	13'-0"	13'-6"	14'-0"
4	261	236	225	190	179	164	151	139
4.5	317	287	261	238	218	200	184	170
5	377	342	311	284	260	238	219	202
5.5	436	398	362	330	303	278	254	238
6	400	400	400	380	348	319	294	272
6.5	400	400	400	400	393	361	333	307